Method Statement

Ref. #: DCP00/08-0205-A-2026



DonProof LE

(High quality polyester reinforced, UV resistant, PVC roofing membrane)

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Section A: General Comments

General Notes:

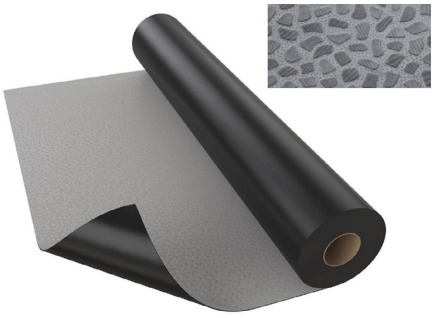
The information below is a detailed overview of the application of DCP's **DonProof LE** waterproofing membrane and should be read in conjunction with the relevant technical data sheet prior to application. All DCP Products should be applied by experienced specialist contractors.

All the points below assume the correct preparation of the relevant surface.

- Guidelines and recommendations as described in EN 13956 should be followed.
- Manufacturer's installation instructions must be always followed.
- **DonProof LE** must be installed by experienced and qualified personnel.
- The substrate onto which **DonProof LE** is to be installed must be sufficiently cured and dry.
- Any poorly attached materials, sharp objects or contaminations must be removed from the substrate prior to installation.
- A separation layer must be applied under all conditions to which the membrane is exposed during its service life, to separate incompatible materials and prevent damage.
- Where **DonProof LE** is installed over old bitumen roofs, timber or rubber, a migration barrier of 140 g/m² polypropylene fleece is required to prevent plasticizer loss from reaction with bitumen.
- Where **DonProof LE** is installed over EPS/XPS insulation, a migration barrier of 120 g/m² glass fleece is required.
- Where **DonProof LE** is installed over smooth concrete or lightweight concrete, a sliding/leveling layer of 300 g/m² polypropylene fleece is required to reduce friction and mask substrate irregularities.
- Fastener spacing along the longitudinal sheet edges must be determined by the wind load calculation for the roof.
- Welding of sheets must always be completed before mechanical fixing is carried out.
- **DonProof LE** must not be installed loose, other than in vacuum roof systems, and must not be installed with gravel ballast.
- Welding must be carried out using an automatic hot air welding machine, in a well-ventilated area.

System Products:

Separation/protection layers below the Membrane: **Geotex** (weighing no less than 300 g/m²).
Waterproofing Membrane: **DonProof LE.**





Tools and Equipment:

It is suggested that the following list of equipment are adopted as a minimum requirement

Personal protection	:	*Protective overalls*
	:	*Goggles or a face mask*
	:	*Good quality gloves*
	:	*Safety shoes*
	:	*Safety helmet*



Application equipment	:	*Automatic hot-air Welding Machine (Fig.1)*
	:	*Utility knife with blade (Fig.2)*
	:	*Pressure roller (Fig.3)*
	:	*Nail gun + Nails (Fig.4)*
	:	*Measuring & marking tools (Fig.5)*



Fig.1: Automatic hot-air WM



Fig.2: Utility knife with blade



Fig.3: Pressure roller



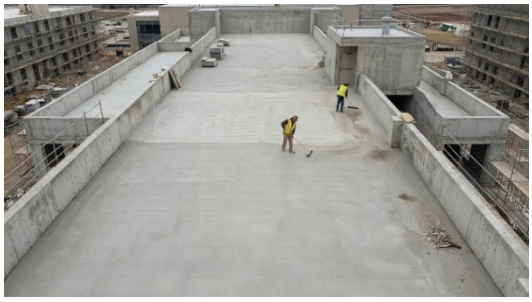
Fig.4: Nail Gun + Nails



Fig.5: Measuring & marking tools

Section B: Application

1.0 Substrate Preparation



1.1 Roofs onto which **DonProof LE** is to be installed must be sufficiently cured and dry.

1.2 Remove any poorly attached materials, sharp objects, or contaminations from the substrate before installation.

1.3 The substrate must not have sharp edges, protruding fasteners, or objects that could puncture or damage the membrane or any underlying separation/vapour control layer.

1.4 Roof pitch is classified as:

- Flat roof: roof angle < 10°
- Sloping roof: roof angle ≥ 10° Always follow local guidelines regarding roof angles and minimum falls.

1.5 A continuous, correctly designed vapour control layer must be installed where required by the building's internal/external moisture load calculation, with all joints, laps, and terminations made airtight (overlap, taped, mastic-bonded, or welded joints). Welded joints give the most reliable airtight result.

2.0 Separation/Protection Layers below the Membrane

2.1 A separation layer must be installed under all conditions to which the **DonProof LE** membrane is exposed during its service life, to separate incompatible materials and prevent damage. The categories of layer are:



- Migration barrier — laid between two layers to prevent chemical interaction between them (specifically, to prevent plasticizer migration between the membrane and an incompatible substrate).
- Sliding/leveling layer — laid between two layers to reduce friction and mask irregularities in the substrate.

2.2 Migration barrier requirements:

Underlying substrate	Migration barrier required
Old bitumen roofs, timber, rubber	140 g/m² polypropylene fleece (**Geotex**)
EPS/XPS insulation	120 g/m² glass fleece

2.3 Sliding/leveling layer requirement:

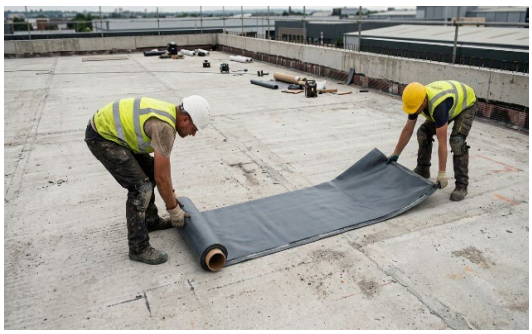
Underlying substrate	Sliding/leveling layer required
Smooth concrete or lightweight concrete	300 g/m² polypropylene fleece (**Geotex**)

2.4 Separation layers are laid loose, with an overlap of approximately 120 mm between adjacent strips.

3.0 Installation of DonProof LE

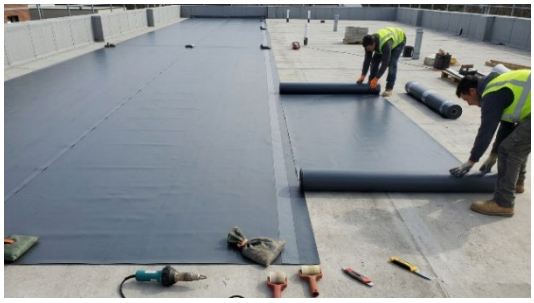
3.1 Preparation

3.1.1 Confirm that substrate preparation (Section 1.0) and all required separation/migration/sliding layers (Section 2.0) are complete before rolling out the membrane.



3.1.2 Plan the roll layout and direction in advance, taking into account the roof geometry, fall direction, and the wind load zones, to minimize the number of joints and ensure correct overlap orientation.

3.1.3 Inspect membrane rolls for transport or storage damage before installation. Rolls must have been stored per the **DonProof LE** storage requirements (original packaging, away from direct sunlight, cool/dry/frost-free conditions) until use.



3.1.4 Confirm the correct fastener type and length are available for the substrate, as determined by the wind load calculation

3.2 Roll-Out, Alignment and Mechanical Fixing

3.2.1 Roll out the first membrane sheet in the intended direction.

3.2.2 Straighten and tighten the sheet to eliminate wrinkles.

3.2.3 Place 2 – 3 fixings at both ends of the sheet to hold it in position.



3.2.4 Install fasteners along both longitudinal edges of the first sheet, at the interval determined by the wind load calculation for the roof.

3.2.5 Roll out and stretch the next sheet, fixing it at both ends, before welding it to the previous sheet (see Section 3.3 — Welding).

3.2.6 After welding, install fasteners along the loose longitudinal edge of the new sheet, following the pre-marked guideline on the sheet.

3.2.7 Overlap and fastener position:

Standard overlap: 120 mm, with fasteners positioned approximately 37 mm from the sheet edge.
2 m-wide sheets: overlap increased to 130 mm.

3.2.8 Fastener edge distance from element edge, when fixing into concrete elements:

Concrete: fixings must never be placed closer than 50 mm from the element edge.
Lightweight concrete: fixings must never be placed closer than 100 mm from the element edge.

3.2.9 **Rule:** welding of the overlap must always be completed before mechanical fixing of the loose longitudinal edge is carried out.

3.2.10 **DonProof LE** must not be installed loose, except in vacuum roof systems, and must never be installed with gravel ballast.

3.2.11 To achieve even stretch across the sheet, install fasteners working outward from the center of the sheet toward the sides, or install end-to-end while pulling ("kicking") the membrane taut immediately before each fastener is placed.

3.2.12 Rectangular washers/fastener plates should be installed with their long side parallel to the longitudinal edge of the sheet.

3.2.13 As a general guide, fastener spacing along the sheet edge should not be less than 200 mm nor more than 1000 mm; for 2 m-wide sheets, spacing must not exceed 530 mm — always subject to the project-specific wind load calculation.

3.3 Welding of Overlaps and Seam Testing

3.3.1 Weld all overlaps using an automatic hot air welding machine, 40 mm nozzle.

3.3.2 Welding must be carried out in a well-ventilated area.

3.3.3 Rule: welding of the overlap must always be completed before mechanical fixing of the loose longitudinal edge is carried out (restated from Section 3.2).



3.3.4 Both sides of the overlap are heated to melting point (approximately 170°C for PVC) and pressed together with the machine's pressure roller. The PVC from both sides melts homogeneously together to form the joint.

3.3.5 A correct weld shows good PVC melt-out along the seam:

If the heat is too high or the machine speed too slow, the membrane will burn/char.

If the heat is too low or the machine speed too fast, there will be no melt-out and the weld can be easily separated.

3.3.6 Test-weld to determine the correct temperature and speed settings before starting production welding, and re-check settings whenever outdoor temperature, humidity, or membrane thickness changes.



3.3.7 For T-joints, additional pressure must be applied by hand or with a pressure roller immediately after the machine passes over the joint, to avoid an open channel (leak path) at the intersection.

3.3.8 Spot/tack welding of the membrane in place (to hold it before machine welding) is prohibited, except in the very limited internal-pipe-collar detailing case described under pipe penetration detailing (to be covered under Section 4.0 — Details and Termination Work).

3.3.9 Test welds at the start of welding and at intervals of every 200 linear metres of welding thereafter.

3.3.10 **Manual test with seam probe:** place the point of the seam probe against the edge of the welded seam and pull it along the seam while applying light pressure. Where the probe penetrates an unwelded area ("fish mouth"), pull the membrane apart until sound weld is reached, then re-weld using the hand-held welding tool.

3.3.11 **Tear/peel test:** cut a weld sample from the machine-welded seam; the weld must not separate at the seam itself but tear in the membrane weave. Where a sample is cut for testing, cover the location with a "Quality Inspection" patch.

3.3.12 **Vacuum test** (for exposed detail welds such as T-joints):

Equipment required: vacuum pump with manometer and regulator, transparent vacuum bell, leak-indicator liquid (e.g. soapy water).

Procedure: spray indicator liquid onto the joint, place the vacuum bell over the test area, draw a vacuum for 2 - 3 seconds. Any leak will show as bubbles. Mark and re-weld any weak points found.

3.3.13 **Water pressure test:** only suitable where any resulting leak would not damage the structure (e.g. an unoccupied concrete/steel deck before covering); not suitable for occupied buildings or roofs housing an operational business. Not to be used at or below freezing temperatures.

3.4 Vacuum Roof Systems



3.4.1 In a vacuum roof system, DonProof LE is laid loosely over an airtight, load-bearing substrate and is anchored and made airtight only at the roof perimeter, at the parapet, and around penetrations — no field fasteners are used.

3.4.2 Wind flow over the roof surface creates negative pressure (suction) in the air layer between the membrane and the airtight substrate. This suction holds the membrane in place without mechanical fixing; the stronger the wind, the more firmly the membrane is held down.

3.4.3 The substrate and the underlying vapour control layer must be airtight and load-bearing. The roof's suitability for vacuum fixing must be assessed and confirmed by DCP Technical Department before this system is selected.

3.4.4 The membrane is supplied and installed in wide rolls (2 m or greater width, or as prefabricated sheets) to minimize the number of field joints.

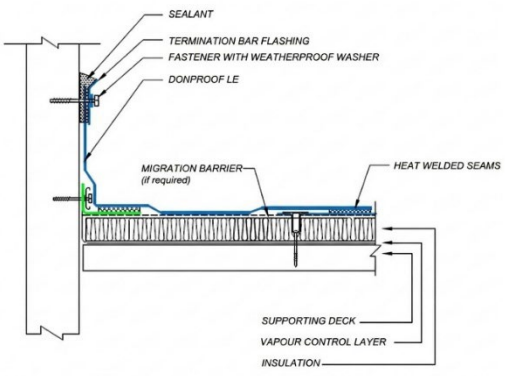
3.4.5 All penetrations, rainwater outlets, and the connection to the parapet must be made fully airtight, using air-sealing tape installed under a linear load-distributing steel bar.

3.4.6 Vacuum vents (one-way valves that release air but do not admit it) must be installed where negative pressure is expected to be greatest — typically the corner and perimeter zones — to relieve overpressure and vent any air leaks from imperfectly sealed areas. The number and position of vents must be determined by DCP Technical Department in consultation with the project's wind exposure data.

3.4.7 Field seams and detail welds in a vacuum system are welded using the same hot air welding procedure and seam testing methods described in Section 3.3.

4.0 Details and Termination of Work

4.1 Flashings — General



4.1.1 Flashings are made using steel strips mechanically fastened to resist horizontal forces, fixed at ≤ 200 mm centers.

4.1.2 Details must be windproof and designed to prevent water from running behind or under the membrane.

4.1.3 The flashing-to-field-membrane connection must be no more than 200 mm (20 cm) from the upstand.

4.1.4 Sealants or mastics may be used to seal profiles.

4.1.5 Perimeter fixings (around upstands and penetrations) require approximately 5 fasteners per linear metre, using steel bars or laminated metal strips.

4.1.6 In valleys with a slope change of ≥ 3°, and at wall connections, install fixings at ≤ 200 mm centers.

4.2 Flashing to High Parapet

4.2.1 For parapets taller than 500 mm, the flashing membrane must be fully bonded to the vertical substrate using contact adhesive and mechanically fixed at the top.

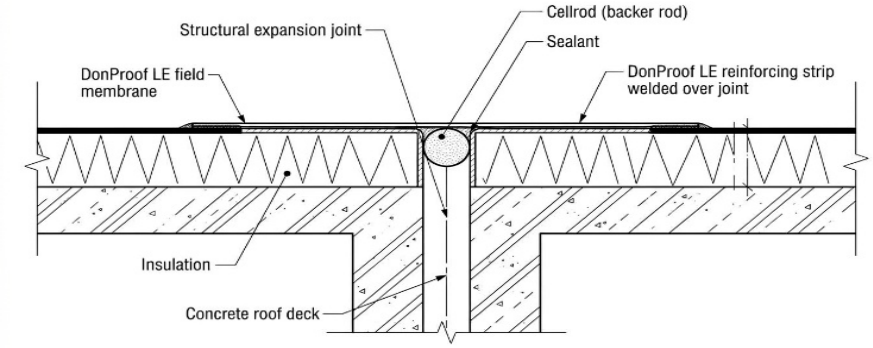
4.2.2 As an alternative to full bonding, the membrane flashing can be loose-laid with an intermediate mechanical fastener row, executed either as a two-part covered connection or as a single membrane strip with an intermediate fastener and cover strip.

4.2.3 Maximum vertical spacing between fastener rows is 500 mm.

4.3 Expansion Joints

4.3.1 The layers in the roof structure must have joints wherever there are joints in the underlying supporting structure of the building.

4.3.2 Structural and expansion joints should normally be designed as high points. They must not run through roof openings, and must not be closer than 500 mm to such openings.



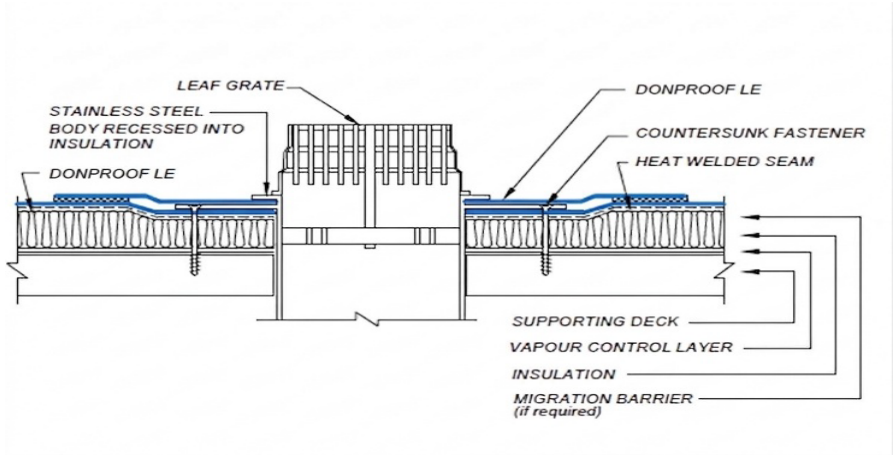
4.4 Rainwater Drainage Outlets

4.4.1 Use prefabricated drains with factory-welded membrane collars, or drains equipped with clamping rings. Drains should be positioned at least 300 mm away from walls, parapets, joints, or other structural elements.

4.4.2 Each roof section separated by an expansion joint must have its own dedicated drain. Follow local/national regulations for drain quantity and sizing.

4.4.3 Where outlets include clamping flanges and sealing gaskets, the membrane is clamped between the top and bottom gaskets.

4.4.4 The drain must be installed at the lowest point of the roof; recess the thermal insulation to form a shallow sump if needed for proper drainage.

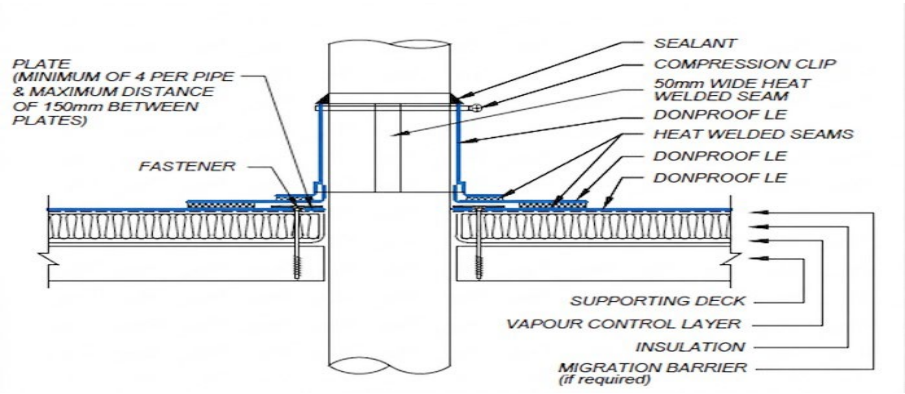


4.4.5 The drainage system must be sized per national requirements, with provision for emergency overflow.

4.5 Pipe Penetrations

4.5.1 Pipe penetrations are normally sealed using prefabricated pipe details, or from a flange (approx. 500 x 500 mm) and a collar made from homogeneous membrane.

4.5.2 Install four fasteners in the field membrane around the base of the pipe. Cut a hole in the middle of the flange, approximately 60 mm smaller than the pipe diameter.



4.5.3 Using a hot air hand tool, heat the flange to make it pliable, then pull it over the protruding pipe (this adjustment can be made on a loose pipe of the same diameter if easier).

4.5.4 The membrane collar, being of smaller diameter than the pipe, will stand up a few centimetres around the pipe. Weld the collar to the field membrane.

4.5.5 Install the flashing strip and hot-air weld it to the upstanding collar material.

5.0 Cleaning

5.1 Tools and equipment (welding machine, nozzle, pressure roller) can be cleaned with water.

5.2 Clean the welding machine, nozzle pieces, and hand tools regularly using a wire brush and clean water to remove PVC residue and prevent contamination of subsequent welds.

5.3 The underside of the succeeding sheet and the upper side of the previously-laid membrane must be clean, dry, and free from any contamination before welding.

Section C: Cautions

Health and safety

- **DonProof LE** should be welded in a well-ventilated area. In closed areas, use forced ventilation and wear approved respirators when necessary.
- Hot air welding of PVC generates acidic fumes; avoid breathing welding fumes and wear appropriate respiratory protection during welding operations.
- **DonProof LE** should not come into contact with skin and eyes. In case of eye contact, wash immediately with plenty of water and seek medical advice promptly.

Fire:

DonProof LE is nonflammable.

For further information on refer to the Material Safety Data Sheet.

Section D: Approval and Variations

This method statement is offered by DCP as a 'standard proposal' for the application of **DonProof LE**. It remains the responsibility of the Engineer to determine the correct method for any given application. Where alternative methods are to be used, these must be submitted to DCP for approval, in writing, prior to commencement of any work. DCP will not accept responsibility or liability for variations to the above method statement under any other condition.